

Behrooz Moosavi Ramezanzadeh

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RESEARCH INTERESTS

- **Applied Analysis**
 - **Calculus of Variations:** Regularity Theory, Relaxation, Homogenization.
 - **PDE:** Nonlinear Elliptic PDE, Viscosity Solutions, Regularity of Solutions.

EDUCATION

University of Pittsburgh

Pittsburgh, PA, USA

- *M.S. in Mathematics* *expected May 2027*
Thesis: *Partial Differential Equations and Stochastic Tug-of-War Games.*
Advisor: [Juan J. Manfredi](#).
GPA: 3.96/4.00.
- *Ph.D. student in Economics* *Aug. 2023 – present*
Advisors: [Jordan Awan](#) and [Arie Beresteanu](#).
GPA: 3.84/4.00.
- *M.A. in Economics* *conferred Aug. 2025*

Sharif University of Technology

Tehran, Iran

B.S. in Mathematical Sciences *Aug. 2016 – Jan. 2022*
GPA: 3.78/4.00 — ranked 4th in a graduating class of 80.

PUBLICATIONS AND PREPRINTS

Applied Mathematics

- A Projected Tug-of-War Game for the Regularized p -Laplacian** *2026*
Single-authored. *Journal of Elliptic and Parabolic Equations.* [DOI] [arXiv:2605.01218]
- An Asymptotic Mean Value Characterization for the Regularized p -Laplacian** *2026*
Single-authored. Preprint. [arXiv:2607.02910]
- Optimal Control of an Epidemic with Intervention Design** *2025*
Single-authored. Revise and resubmit. [arXiv:2503.22928]
- Extended Set Difference: An Inverse Operation of Minkowski Summation** *2024*
With [Arie Beresteanu](#). Revise and resubmit. [arXiv:2412.19779]

Econometrics/Statistics

- Partial Identification with Auxiliary Moment Restrictions** *2026*
With [Arie Beresteanu](#). Preprint. [arXiv:2607.21807]
- Public Good Provision under Locally Private Signals** *2026*
With [Jordan Awan](#). Preprint. [arXiv:2606.24013]
- Bounds for Restricted Selections of Random Sets** *2025*
With [Arie Beresteanu](#). Preprint. [arXiv:2512.04539]

RESEARCH EXPERIENCE

Research — Calculus of Variations

Spring 2026 – Present

Carnegie Mellon University. Advisor: [Irene Fonseca](#).

- Study relaxation, lower semicontinuity, and $\mathcal{A}$ -quasiconvexity for variational problems under variable-coefficient PDE constraints.
- Develop a moving-kernel approach to variable-coefficient $\mathcal{A}$ -free oscillations, with the goal of obtaining necessity results without pseudodifferential machinery and extending global relaxation methods to the variable-coefficient setting.

M.S. Thesis Research — Nonlinear PDE and Stochastic Games

Fall 2025 – present

University of Pittsburgh. Advisor: [Juan J. Manfredi](#).

- Study nonlinear elliptic equations through stochastic tug-of-war games and asymptotic mean value characterizations.
- Analyze the regularized p -Laplacian and its game-theoretic interpretation in terms of value functions of discrete games.
- Construct projected tug-of-war games and derive the corresponding asymptotic mean value formulas, obtaining equivalence with viscosity solutions.

HONORS AND AWARDS

Gold Medal, Iranian National Mathematical Olympiad

2014

National competition, Iran.

Ph.D. Preliminary Examinations, University of Pittsburgh

2026

Passed the preliminary examinations in Analysis (60/60) and Linear Algebra (57.7/60) on the first attempt.

Distinguished Student, Sharif University of Technology

2021

Ranked 4th in a graduating class of 80, Department of Mathematical Sciences.

Iranian National University Entrance Examination

2016

Ranked 48th among approximately 650,000 participants in the Mathematics and Physics track.

TALKS, CONFERENCES, AND SUMMER SCHOOLS

Topics at the Interface of Partial Differential Equations and Probability

2026

Summer school participant, Department of Mathematical Sciences, Carnegie Mellon University.

Midwest Econometrics Group Annual Meeting

2025

Selected contributed talk, University of Illinois Urbana–Champaign. [\[Slides\]](#)

TEACHING EXPERIENCE

University of Pittsburgh

2022 – 2026

Teaching Assistant: Graduate Stochastic Processes (Fall 2022); Probability & Statistics I (Summer 2023); Intermediate Microeconomics (Fall 2024); Applied Econometrics I, II (Spring, Fall 2025); Introduction to Microeconomic Theory (Spring 2026).

Sharif University of Technology

2017 – 2022

Instructor: Introduction to Linear Algebra for Engineers (Spring 2019) — sole responsibility for lectures, problem sets, and assessment.

Teaching Assistant: Abstract Algebra I (Spring 2018); Topology I (Spring 2021); Elementary Measure Theory (Spring 2022). *Volunteer tutor*, Yarigaran Center, for child laborers (2017–2018).

SELECTED ADVANCED COURSEWORK

Carnegie Mellon Univ.	Univ. of Pittsburgh	Sharif Univ. of Technology
Calculus of Variations	PDE I, II	Real Analysis I, II
Functional Analysis	Analysis II (Sobolev Spaces)	Complex Analysis
Measure and Integration	Analysis III (Harmonic)	Topology
Advanced Optimal Transport	Differential Geometry I, II	Abstract Algebra I
Stat Optimal Transport I, II	Probability Theory I	Dynamical Systems
	Applied Stochastic Methods	Numerical Analysis
	Mathematics of Finance I, II	Linear Algebra

Carnegie Mellon courses were taken through cross-registration.

TECHNICAL SKILLS AND LANGUAGES

- **Computing:** Python, MATLAB, Julia, R, C++, Lean 4, $\text{\LaTeX}$; author of the CRAN package [LangevinFlow](#).
- **Languages:** Persian (native), English (fluent).